# Siraj J. Lyons

Curriculum Vitae

Email: siraj.lyons@louisville.edu

Website: koffeinhaltigt-lejon.github.io

---

## Education

University of Louisville, PhD — 2022 – *Present*
Psychological Science
Advisor: Dr. Brendan Depue
*Dissertation: Brainstem nuclei in bipolar disorder: A morphometric and resting-state entropy investigation of the ascending arousal network*

Northern Michigan University, M.S. — 2020 – 2022
Psychological Science
Advisor: Dr. Lin Fang
*Thesis*: *Electrocortical changes following attention bias modification in a sample of high-bipolar trait individuals*

University of Toledo, B.A. — 2016 – 2020
Psychology

---

## Research Experience

**Graduate Student** — **2022 - *Present***
PI: Brendan E. Depue, PhD
Psychological Science
University of Louisville, Louisville, KY

**Graduate Student** — **2020 – 2022**
PI: Lin Fang, PhD
Department of Psychological Science
Northern Michigan University, Marquette, MI

**Research Assistant** — **2018 – 2020**
PIs: John D. Jasper, PhD; Wesley Bullock, PhD; Matthew Tull, PhD; Kim Gratz, PhD
Department of Psychology
University of Toledo

## Manuscripts

** denotes undergraduate student*; *# denotes co-first author*

Foley, F., **Lyons, S. J.**, Cook, O., & Depue, B. E. (2026). Investigating the Functional Neural Network Dynamics of Cognitive-Affective Action Planning during Threat Processing. *Frontiers in Psychology*.

Lawrence, J. M., Foote, I. F., Breunig, S., Schaffer, L. S., **Lyons, S.**, Abramowitz, S., Levin, M. G., Damrauer, S. M., Mallard, T. T., Grotzinger, A. D. (2026). Shared genetic liability across systems of psychiatric and physical illness. *Nature Communications*. https://doi.org/10.1038/s41467-026-69218-1

**Lyons, S.**, Beck, I., & Depue, B. E. (2025). Depression is marked by differences in structural covariance between deep brain nuclei and sensorimotor cortex. *NeuroImage*, 310. https://doi.org/10.1016/j.neuroimage.2025.121127

**Lyons, S.** & Depue, B. E. (2023). Not all bad decisions are alike: Approach and avoidant bad decisions are associated with distinct network organization. *Frontiers in Neuroscience*, *17*. Article number: 1249008. https://doi.org/10.3389/fnins.2023.1249008.

### *Submitted*

**Lyons, S. J.**, Foley, J., Cook, O., & Depue, B. E. (*Under Review*). Bioinformed idiographic symptom networks to identify neurobiological predictors of affective-state transitions in bipolar disorder.

**Lyons, S. J.**, & Depue, B. E. (*Under Review*). Sensorimotor inference as an adjunctive model of bipolar disorder.

### *In preparation*

**Lyons, S. J.,** Saki, J., Reid, A., & Jasper, J.D. (*In Prep*). Effects of leftward unilateral gaze on Iowa Gambling Task performance across the hypomanic personality spectrum depends on handedness. *Target Journal: Journal of Behavioral Decision Making*. IF: 1.8

**Lyons, S. J.**, Kolks, S.*, Komula, B.*, Foley, J. F., Depue, B. E. (*In Prep*). Morphometric and functional relationships between the ascending arousal network and heart rate variability.

**Lyons, S. J.**, Foley, J. F., Gruner, S.*, & Fang, L. (*In Prep*). Personality clusters predicting attention bias toward monetary images: A data-driven approach.

## Grants & Awards

**Dean's Citation Award** 2024
University of Louisville
Award presented to a graduating Master's or Doctoral student whose work excelled in research, teaching, or service beyond a high-grade point average. Award given for Masters.

**Excellence in Research (Junior Level)** 2024
University of Louisville: Department of Psychological and Brain Sciences
Department award recognizing research trainees who have excelled in research who have yet to enter doctoral candidacy

**Travel Award** 2023
Granting Institution: University of Louisville
$350 intramural award for presenting research at a national conference

**University Fellowship** 2022 – 2024
University of Louisville

**Dean Research Award** 2022
Granting Institution: Northern Michigan University
$500 intramural award funding summer research

**Excellence in Education** 2021
Granting Institution: Northern Michigan University
$1500 intramural grant funding summer research
PI: Dr. Lin Fang

## Supervision

Samantha Kolks 2026 – *Present*
Undergraduate Research Assistant
University of Louisville

Bruce Komula 2026 – *Present*
Undergraduate Research Assistant
University of Louisville

Jake Cox 2025
Undergraduate Honors Thesis
University of Louisville

Seth Gruner 2023 – *Present*

Undergraduate Research Assistant
University of Louisville

Sasha Domenech (Pham) 2023 – 2024
Undergraduate Research Assistant
University of Louisville

Jackson Bower 2023
Undergraduate Research Assistant
University of Louisville

Zachary Irvin 2022 - 2023
Undergraduate Research Assistant
University of Louisville

John Foley 2020 – 2022
Undergraduate Research Assistant
Northern Michigan University

Haley Gaboury 2020 – 2022
Undergraduate Research Assistant
Northern Michigan University

Alex Reid 2019 – 2020
Undergraduate Research Assistant
University of Toledo

---

## Presentations

**Lyons, S.** (September, 2025). Ascending arousal network morphometry and Big 5 personality traits: A data-driven analysis. University of Louisville – Program Colloquium.

**Lyons, S.** (April, 2025). Deep-brain nuclei structural covariance with sensorimotor cortex is associated with major depressive disorder. *Invited Talk*. Northern Michigan University. **CANCELLED DUE TO FUNDING.**

**Lyons, S.** (February, 2025). Deep-brain nuclei and Major Depressive Disorder: What's the connection? University of Louisville – Program Colloquium.

**Lyons, S.** (February, 2024). A brief survey of reward in the brain: Future Avenues. University of Louisville – Program Colloquium.

---

## Conference Posters

** denotes undergraduate student*

**Lyons, S.,** Beck, I., & Depue, B. (March – April, 2025). Structural co-dependence between deep brain nuclei and sensorimotor cortex differs between people with depression and unaffected individuals.

**Lyons, S.**, Irvin, Z*., & Depue, B. (November, 2023). Frontoparietal network stability predicts attentional control, not cognitive flexibility. Society for Neuroscience.

Cook, O., **Lyons, S.**, Depue, B. (March, 2023). Cognitive and Motor Neural Network Interaction during Threat Response. Cognitive Neuroscience Society.

**Lyons, S.** & Depue, B. (March, 2023). Behavioral approach is linked to increased neural communication between frontostriatal and insular regions during risky decision-making. Cognitive Neuroscience Society.

**Lyons, S.,** Carlson, J. M., & Fang, L. (November, 2022). Frontal and striatal volumes at baseline predict changes in anxiety symptomology following attention bias modification. Society for Neuroscience.

Gentry, L., **Lyons, S.**, Kassel, D., Davis, B., Kwak, B., Carlson, J. M., & Fang, L. (November, 2022). Reduced intrinsic connectivity between the dorsal anterior cingulate and medial prefrontal cortices predicts anxiolytic benefits of attention bias modification training in high trait anxious individuals. Society for Neuroscience.

Hauler, A., Foley, J., Kassel, D., **Lyons, S.**, Caughtry-Carpenter, C., Fang, L., Carlson, J. (September-October, 2022). Reliability of Eye Tracking Based Measures of Attentional Bias in the Scrambled Sentence Task. Society for Psychophysiological Research.

Kassel, D., Hauler, A., **Lyons, S.**, Gaboury, H., Girard, C., Carlson, J. M., & Fang, L. (April, 2022). Shared and Unique Patterns of Resting-State Functional Connectivity Linked to Attentional and Interpretation Biases. Cognitive Neuroscience Society.

**Lyons, S.**, Gaboury, H.*, Foley, J.*, Delgado, X., Hauler, A., Carlson, J. M., & Fang, L. (April 2022). Changes in attention and reward-sensitivity ERPs following attention training in a sample of bipolar disorder susceptible individuals. Cognitive Neuroscience Society.

**Lyons, S.**, Gilbertson, H., Girard, C., Carlson, J. M., & Fang, L. (November, 2021). Prefrontal cortex based intrinsic functional connectivity is related to electrocortical measures of conflict monitoring. Society for Neuroscience.

---

## Service

**Preliminary Exam Q&A** 2026
Served as a panelist for a program Q&A session for pre-candidacy PhD students

**Pursuing Liberty, Unity, and Support (+) Committee** 2025 – 2026
Graduate Student Member of Reports Subcommittee.

**Graduate Network in A&S (GNAS) Representative, University of Louisville** 2025 – 2026
Chair of the Academic and Grants committee

**Peer Mentor, University of Louisville** 2024 - 2026
Point of contact for a 1st year graduate student in the Experimental Psychology program.

---

# Teaching Experience

**Graduate Teaching Assistant**, University of Louisville 2024 – Present

- PSYC 305: Brain and Behavior; *N* = 100
  - Assisted in developing instructional material, quizzes, and exams.
- PSYC 381: Cognitive Neuroscience; *N* = 60
  - Assisted instructor of record with grading exams and short-form essays. Gave lecture on *Emotion Neuroscience* under supervision of instructor of record.
- PSYC 301: Psychological Statistics; *N* = 25 per section
  - Developed lab materials to facilitate active practice of core principles and statistical tests found in psychological sciences.
  - Led instruction during lab component
  - Taught Fall 2025 and Spring 2026

**Graduate Teaching Assistant**, Northern Michigan University 2020 – 2022

- PSY 100 – Introduction to Psychology (*N* = 28)
  - Fall 2020 – Spring 2022
  - Lab Instructor
- PSY 250 – Brain and Behavior (*N* = 20)
  - Fall 2020 – Spring 2022
  - Lab Instructor
  - Trained graduate teaching assistant on teaching protocol Fall 2022

**Undergraduate Teaching Assistant**, University of Toledo 2020

- Research Methods (*N* = 13)
  - Assistant to primary instructor and graduate teaching assistant for a course in research methods. Graded quizzes and assisted in reviewing rough drafts of an APA-style report. Additional duties included holding office hours.

## Coding (skill-level)

**R** (Proficient)
**MATLAB** (Proficient)
**Bash** (Proficient)
**LaTeX** (Beginner)
**Python** (Beginner)